

• LIVE ON VERCEL

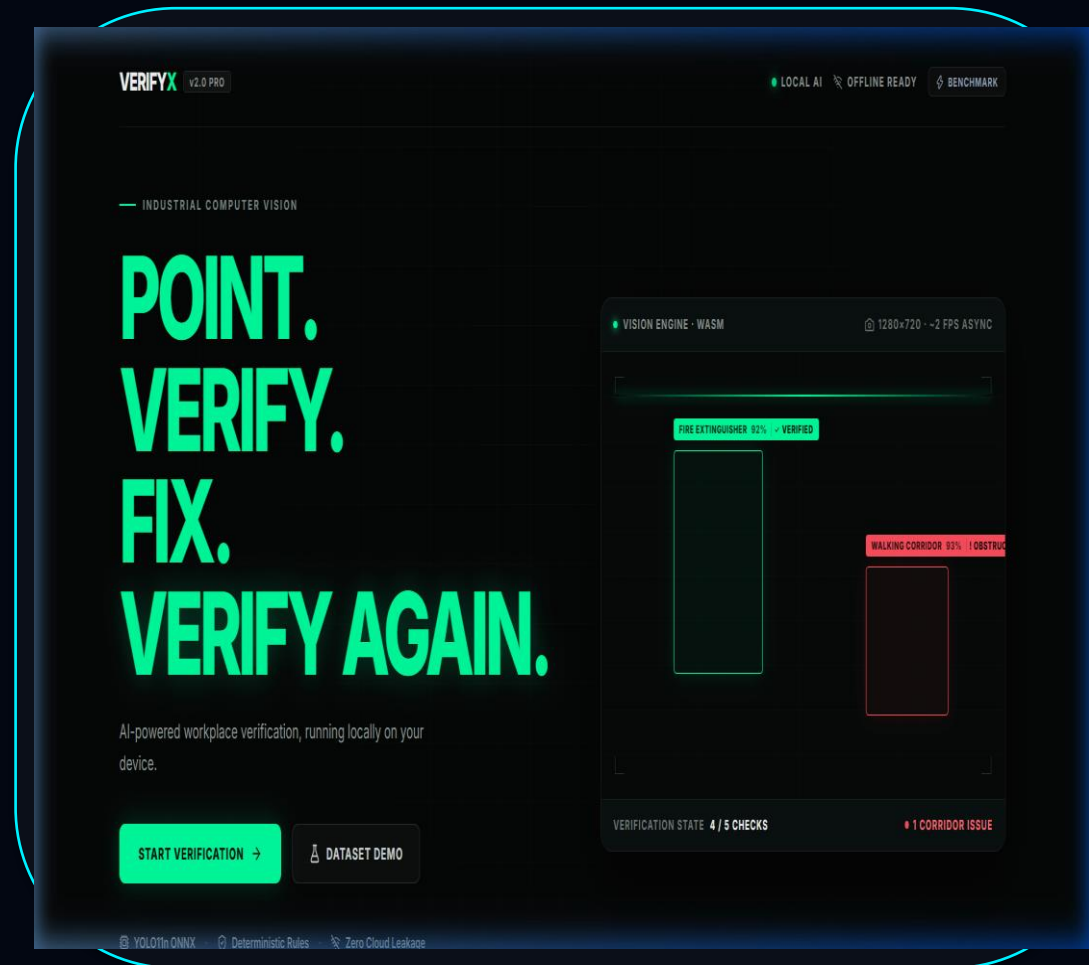
VERIFYX

"Verify Before You Trust."

AI-powered verification for detecting suspicious or potentially non-compliant physical workplace conditions.

A phone-first, in-browser AI verification engine. Reduces visual inspection ambiguity through automated object detection, deterministic spatial rule evaluation, and closed-loop audit reports.

24/24 UNIT TESTS PASS • DEVICE-LOCAL PROCESSING • ASYNC ~2 FPS HUD



"Seeing is no longer believing."

The modern verification challenge across visual environments

CONTENT —> UNKNOWN ORIGIN —> MANIPULATION / INSPECTION RISK —> UNCERTAIN TRUST

01 INFORMATION OVERLOAD

Facility managers, safety operators, and auditors encounter high volumes of visual scenarios and documentation daily.

The sheer volume and speed of modern operations can easily outpace manual review capacity.

02 PERCEPTION LIMIT

Human visual inspection can struggle with precise spatial relationships, clearance margins, and subtle hazard proximity.

Subjective human review suffers from fatigue, confirmation bias, and inconsistent evaluations.

03 FRAGMENTED VERIFICATION

Photos, notes, physical checklists, and audit reports frequently live in separate, disconnected workflows.

Without a continuous chain of evidence, verification details are lost between field inspection and sign-off.

"The problem isn't finding content. It's knowing whether to trust it."

"Verification should be a workflow, not a guess."

Transforming visual verification from subjective inspection to structured evidence

BEFORE // THE MANUAL GUESS

Capture —> Inspect —> Guess

- Uncalibrated visual appraisal
- Black-box subjective judgement
- No spatial dimension calculations
- Verbal remediation instructions left unverified
- High variance between individual inspectors

AFTER // THE VERIFYX WORKFLOW

CAPTURE —> ANALYZE —> VERIFY —> EXPLAIN —> REPORT

- Automated local object detection (YOLO11n ONNX)
- Geometric spatial rule evaluation (IoU overlap)
- Explicit explainability formula for every verdict
- Enforced Before/After rescan evidence for remediation
- Structured, timestamped JSON inspection records

PERCEPTION + REASONING = ACTIONABLE DECISION

"VERIFYX turns raw visual input into a structured, explainable decision."

Meet VERIFYX.

An AI-powered verification workflow that turns ambiguous visual inputs into an explainable verification result.

01 LIVE SCAN

Real-time camera feed with in-browser inference reticles throttled at ~2 FPS async.

02 DATASET DEMO

Pre-loaded evaluation suite running 5 standardized workplace safety test cases.

03 VERIFICATION RESULTS

Real-time checklist resolution tallying verified standards (4 / 5 VERIFIED).

04 ISSUE DETECTION

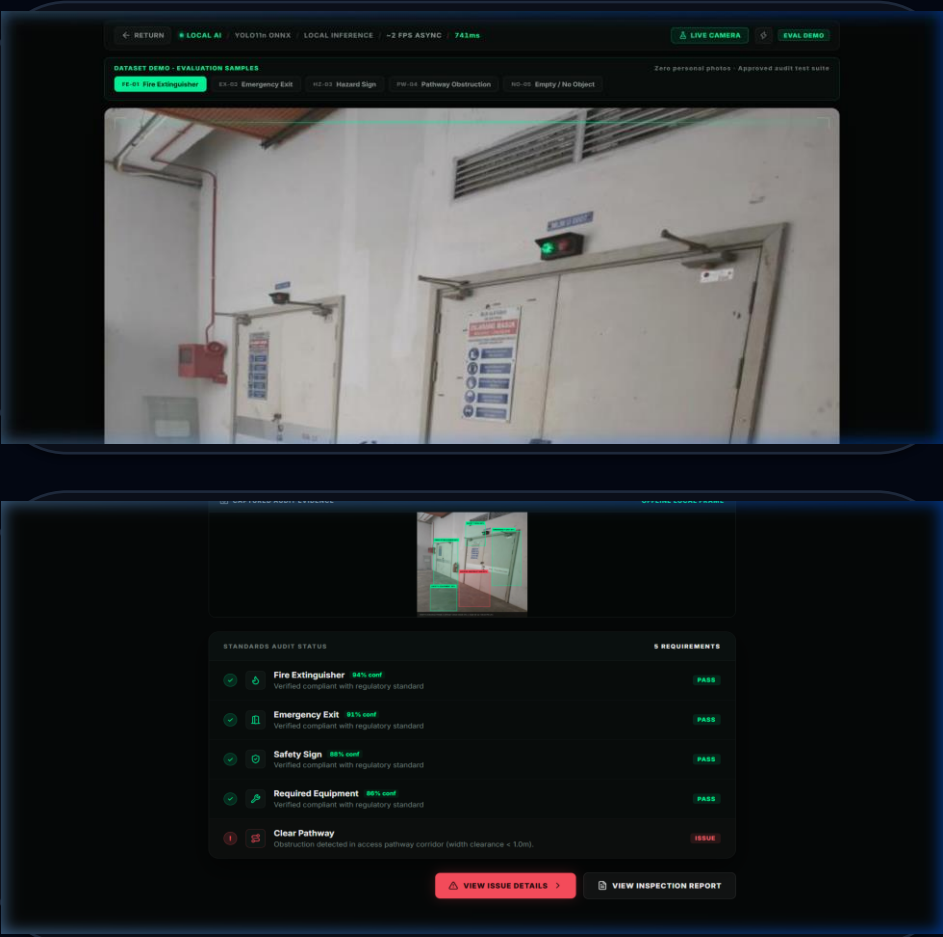
Mathematical spatial rule engine identifying bounding box conflicts & pathway intrusions.

05 RESCAN

Closed-loop remediation workflow: side-by-side Before/After targeting confirming clearance.

06 AUDIT REPORT

Structured inspection log (VX-INSP-2026-XXXX) with timestamps and JSON export.



Inside the VERIFYX verification pipeline.

A 4-tier decoupled architecture combining in-browser neural perception and deterministic spatial logic

USER LAYER

WebRTC Camera Stream (640x640 Canvas) • Dataset Evaluation Suite (5 Test Scenarios)

PROCESSING LAYER

Frame Normalization (RGB, Float32) • Temporal Confirmation (Sliding Window Buffer)

INTELLIGENCE LAYER

Primary: Custom YOLO11n ONNX via WebAssembly (onnxruntime-web 1.30.0)
Fallback: Local TF.js COCO-SSD • Deterministic Simulation Engine
Spatial Engine: Bounding Box Overlap, Size Guardrail (<32px), Egress Pathway Clearance

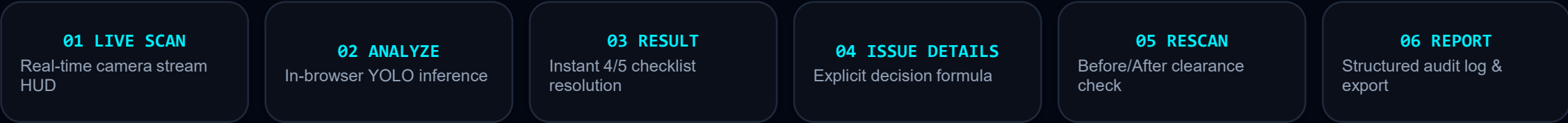
OUTPUT LAYER

Dynamic Reticle HUD • Issue Explainability Matrix • Structured JSON Report (VX-INSP-2026-XXXX)

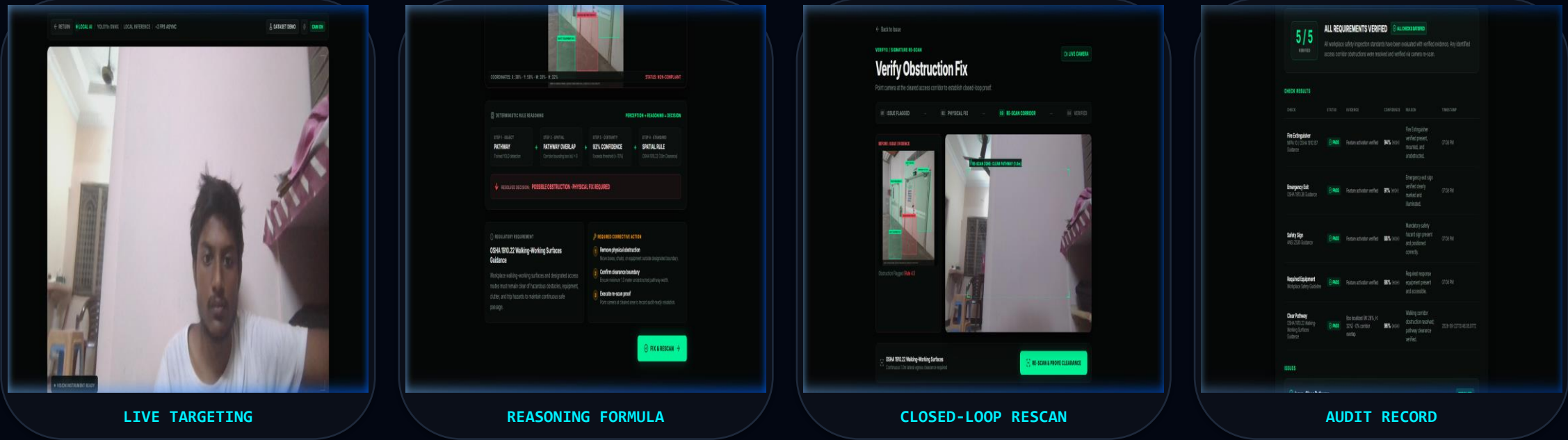
TECHNICAL PRINCIPLE: Camera frames are processed locally in the browser; they are not transmitted to external servers.

From scan to decision in one flow.

The complete closed-loop inspection journey: point, verify, fix, and verify again



EXPLAINABLE FORMULA: DETECTION [Chair @ (x,y,w), Conf: 88%] + SPATIAL RULE [Pathway Overlap > 10%] = POSSIBLE OBSTRUCTION



"VERIFYX doesn't just flag. It explains."

Objective architectural comparison between manual inspection methods and VERIFYX

DIMENSION	MANUAL INSPECTION METHODS	THE VERIFYX WORKFLOW
Tooling	Disconnected: Camera + Paper / Spreadsheets	Single In-Browser Web Application
Analysis	Subjective, uncalibrated visual estimate	Object Detection + Geometric Spatial Rules
Explainability	Intuition-based ('looks blocked')	Explicit Breakdown: Detection + Spatial Rule = Verdict
Remediation	Untracked verbal instruction	Enforced Before/After Rescan Evidence
Audit Trail	Fragmented photos & unlinked notes	Structured JSON with Session ID & Timestamps
Processing	Manual review after the inspection	Real-Time On-Device Assessment
CORE ADVANTAGE: DETECTION + SPATIAL RULE = EXPLAINED VERDICT		

Built. Tested. Deployed.

Demonstrated software engineering discipline with zero mockups and 100% test pass rate

24 / 24

TESTS PASSED (100%)

- 13 Spatial Rule Engine Tests
- 11 Report Lifecycle Tests
- Edge case validation (<32px guardrail)
- Temporal buffer tests
- No fabricated evidence test check
- Dev mock transfer bridge test

BUILD STATUS
PASS (Zero Errors)

MOBILE VIEWPORT
VALIDATED
Touch HUD

DESKTOP VIEWPORT
VALIDATED
Responsive

LIVE SCAN
PASS (~2 FPS Async)

DATASET DEMO
PASS (5 Test Cases)

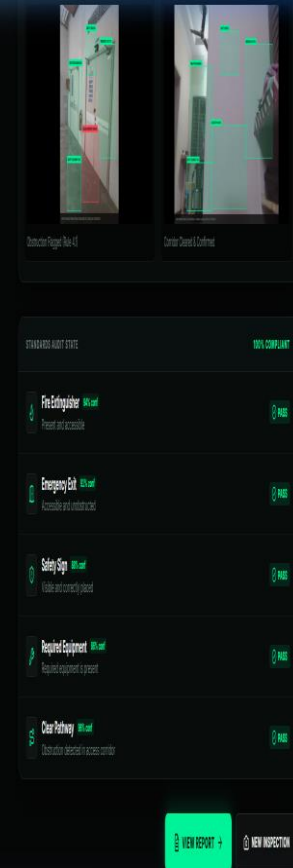
CLOSED RESCAN
PASS (Remediation)

AUDIT REPORT
PASS (JSON Export)

GIT REPOSITORY
Commit afb0952

CLOUD HOSTING
Vercel Edge

LIVE APPLICATION
verifyx.vercel.app



From verification prototype to inspection platform.

Clear separation between current live capabilities and future development horizons

TODAY // LIVE IMPLEMENTED & TESTED

- In-browser verification web application
- YOLO11n ONNX via WebAssembly
- Spatial Rule Engine (clearance & size)
- Tiered fallback pipeline (TF.js / Sim)
- Closed-loop Before/After Rescan
- Structured JSON inspection reports
- Deployed on Vercel Edge Network

NEXT // ROADMAP PLANNED (NOT YET IMPLEMENTED)

- REST & Webhook APIs for facility systems
- Additional inspection categories
- User-configurable spatial rule sets
- Browser extension for desktop verification
- Native iOS / Android wrapper packaging
- Export integrations for safety databases

FUTURE // VISION RESEARCH (LONG-TERM)

- Cryptographic signing for tamper-evident records
- Multi-camera temporal tracking across zones
- Automated compliance export adapters
- Continuous on-device model adaptation
- Universal verification protocol

TARGET USE CASES: Egress corridor clearance, emergency equipment accessibility, facility maintenance records.

DISCLAIMER: VERIFYX is currently a prototype verification rule engine and not a certified safety compliance system.

DON'T JUST CONSUME INFORMATION. VERIFY IT.

VERIFYX

"Verify Before You Trust."

- LIVE WEB APPLICATION
- 24/24 TESTS PASSED
- DEVICE-LOCAL PROCESSING



<https://verifyx.vercel.app>

Runs directly on your smartphone browser.
No app download required.